

Education

- **Daffodil International University** Apr 2022 – Apr 2026
B.Sc. in Computer Science and Engineering (CGPA: 3.93/4.00)
Dhaka, Bangladesh
- **Istanbul Kültür University (Erasmus+ Exchange)** Feb 2025 – Jul 2025
Computer Engineering (CGPA: 3.24/4.00)
Istanbul, Turkey
- **Notre Dame College** Jan 2018 – Jul 2020
Higher Secondary Certificate (HSC) in Science (GPA: 5.00/5.00)
Dhaka, Bangladesh

Publications

• Journals

- **Turjy, R. C. D.**, Bijoy, M. H. I., Paul Bappy, S., Sohel, A., & Uddin, J. (2025). "Feature-optimized hybrid CNN-ML architecture for robust jackfruit disease classification in resource-constrained settings." *Springer Nature Big Data (Q1)*. [Under Review] [\[Proof\]](#)
- Bappy, S. P., Bijoy, M. H. I., **Turjy, R. C. D.**, Tarafdar, S., Nur, F. N., Sohel, A., Hasan, M. Z., Arefin, M. S., Dhar, P. K., & Shimamura, T. (2026). "FedMAM: A Lightweight Federated Learning Framework with DeepMedA3-Net for Privacy-Preserving Cross-Modality Medical Image Analysis." *Medical Image Analysis (Q1)*. [Under Review] [\[Proof\]](#)
- Sohel, A., **Turjy, R. C. D.**, Paul Bappy, S., Assaduzzaman, M., Al Marouf, A., Rokne, J., & Alhajj, R. (2025). "SkinVisualNet: A hybrid deep learning approach leveraging explainable models for identifying Lyme disease from skin rash images." *Machine Learning and Knowledge (MAKE), MDPI*. [Published] [\[Link\]](#)
- Shohel, A., Bijoy, M. H. I., Paul Bappy, S., **Turjy, R. C. D.**, Othman, M., & Samad, M. A. (2025). "JackVisualNet: A fine-tuned hybrid deep learning model for jackfruit disease classification with explainable AI." *PeerJ Computer Science (Q1)*. [Under Major Revision] [\[Proof\]](#)

• Conferences & Datasets

- Bappy, S. P., **Turjy, R. C. D.**, Mondal, A., Bithi, T., Bijoy, M. H. I., & Sohel, A. (2025). "Evaluating deep learning models and explainable AI for mushroom classification." *Proceedings of IDAA 2025, Springer*. [Accepted]
- **Turjy, R. C. D.**, Bappy, S. P., & Sohel, A. (2024). "A hybrid deep learning approach for identifying Lyme disease from skin rash images." *IEEE CS BDC Symposium 2024*. [Published] [\[Link\]](#)
- Sohel, A., Bijoy, M. H. I., **Turjy, R. C. D.**, & Bappy, S. P. (2025). "Jackfruit AgroVision: An extensive dataset for jackfruit disease detection using machine learning." *Mendeley Data*. [Published] [\[Link\]](#)

Experience

- **Teaching Assistant** Oct 2025 – Present
Daffodil International University
Dhaka, Bangladesh
 - Work under the supervision of **Prof. Dr. Md. Fokhray Hossain** (Dean, FSIT), providing technical support for **Machine Learning** projects and **Data Communication** tutorials.
- **Datasoft Manufacturing & Assembly Inc. Ltd.** Jul 2024 – Present
Data Scientist (Part-Time)
Dhaka, Bangladesh
 - **ATM Surveillance:** Achieve **98% accuracy** in theft/fraud detection using **CNN models**; successfully deployed in a **Government Bank of Bangladesh**.
 - **HumanTrue:** Reduce false alarms by **91%** across **600+ stores** via human vs. non-human classification using **3D CNNs, Transformers, and XGBoost** with **MQTT** pipelines.
 - **MoreFish:** Attain **95% accuracy** in NH3 prediction by fusing **100k+ IoT sensor readings** with **LiDAR data** using **XGBoost, SVR, and Transformers**.
 - **Tech Stack:** Python (NumPy, Pandas, Scikit-learn), MQTT, MySQL, Paho, XGBoost, CNNs, Transformers.

Workshops Trainer

- **Daffodil International University (IoT Lab)** Oct 2025; Nov 2024
Workshop Trainer & Speaker Dhaka, Bangladesh
 - Led the 1st and 2nd editions of "Inside IoT" workshops, teaching **end-to-end IoT-ML pipelines, data preprocessing for ML, and ML development for IoT devices.**

Projects

- **VitaCrop AI** [\[GitHub\]](#)
 - Developed an **AI-based crop disease diagnosis system** reducing diagnosis time by **40%**; designed a **low-literacy-friendly UI** and deployed on **PythonAnywhere** for real-time inference.

Leadership & Activities

- **Vice Chair:** IEEE DIU Student Branch WIE AG Mar 2025 – Feb 2026
- **Creative Team Leader:** IEEE DIU Student Branch WIE AG Mar 2024 – Feb 2025
- **Campus Ambassador:** Intel OpenAPI Jan 2024 – Jan 2025

Skills

- **Programming:** Python, C, C++, SQL
- **Machine Learning:** Logistic Regression, SVM, Random Forest, XGBoost, SVR, Decision Trees
- **Deep Learning:** CNN, Transformers, PointNet++, Spectrogram-based Models, XAI
- **Data & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering
- **Systems:** MQTT, MySQL, PythonAnywhere, Real-time Inference Pipelines

Awards & Honors

- **Erasmus+ ICM Mobility Scholarship:** Istanbul Kültür University 2025
- **Best Library User Award:** Daffodil International University 2023
- **75% Academic Excellence Waiver:** B.Sc. in Computer Science and Engineering, DIU 2022 – 2026

References

Professor Dr. Md. Fokhray Hossain

Professor, Dept. of Computer Science and Engineering

Dean, Faculty of Science and Information Technology

Daffodil International University

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